

SRISHTI RAJPUT

Data Scientist · Industry 4.0 & IoT Analytics · AI/ML Engineer
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PROFESSIONAL SUMMARY

Data Scientist and AI/ML Engineer with 2+ years of hands-on industrial experience at Rockwell Automation, where I collected, interpreted, and acted on real-time sensor and SCADA data across food, steel, and fertilizer plants. Now applying that rare operational domain knowledge to build production-grade ML systems — predictive maintenance models, anomaly detectors, and explainable AI pipelines. Dual-diploma certified in Data Science & AI (BIA, 2026). Delivered ₹235M+ in projected financial impact through ML. Uniquely positioned at the intersection of Operational Technology (OT) and Data Science

WORK EXPERIENCE

Field Service Engineer | Rockwell Automation

2023 – 2025 · Pan-India

- **Spearheaded** end-to-end PLC upgrade project at IFFCO (government fertilizer plant, 500+ crore facility), delivering the project **on time and within scope** — independently managing data collection from 40+ field instruments, stakeholder coordination, and system commissioning with zero production downtime.
- **Diagnosed and resolved** critical equipment breakdowns across **food, steel, and process manufacturing plants** under AMC contracts — reducing mean fault-resolution time by **~40%** through systematic root cause analysis and pattern recognition on live PLC data.
- **Analyzed** real-time sensor streams, drive parameters, and SCADA alarm logs across **5+ industrial verticals** — building the domain intuition that now drives my predictive maintenance and IoT anomaly detection models.
- **Engineered** SCADA system upgrades (FactoryTalk) and HMI commissioning for **10+ production lines** — creating structured data acquisition frameworks that mirror real-world industrial data pipelines used in Industry 4.0 ML applications.
- **Collaborated** with cross-functional teams of plant engineers, operations managers, and client stakeholders across **geographically dispersed facilities** — developing the data storytelling and communication skills critical for translating ML insights to non-technical decision-makers.

TECHNICAL SKILLS

- **Languages & Core:** Python (NumPy, Pandas, Scikit-learn, Matplotlib, Plotly), SQL (DDL, DML, Joins, Aggregations), PySpark, Pydantic
- **Machine Learning:** XGBoost, Random Forest, Logistic Regression, K-Means, Ensemble Methods, Hyperparameter Tuning, Cross-Validation, Feature Engineering, Supervised learning, Unsupervised Learning, Bagging, Boosting
- **Industrial & IoT:** PLC (Allen-Bradley), SCADA (FactoryTalk), HMI, Predictive Maintenance, OEE Analytics, Industry 4.0, IIoT Sensor Data, Real-time Data Acquisition
- **Deep Learning & GenAI:** BERT, DistilBERT, LLMs (GPT, Claude, Gemini), LangChain, HuggingFace, LLM Fine-Tuning, Prompt Engineering, ANN, CNN, LSTM
- **XAI & Responsible AI:** SHAP, LIME, AIF360, Fairness Auditing, Demographic Parity, Equal Opportunity, EU AI Act Compliance
- **Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Excel Advanced Analytics
- **Tools & Deployment:** Git/GitHub, Streamlit, Gradio, FastAPI, Docker, Databricks, Jupyter, VS Code, n8n, Google ADK

KEY PROJECTS

FrauHawk — Real-Time Financial Fraud Detection | [GitHub ↗](#)

Python · XGBoost · Random Forest · SHAP · Scikit-learn · Streamlit

- **Built** an XGBoost + Random Forest ensemble fraud detector on **11,000+ real transactions**, achieving **100% precision, 99.6% recall, and ROC-AUC of 1.0** — eliminating virtually all missed fraud cases.
- **Quantified ₹235M+ in projected annual savings** by reducing missed-fraud loss by **85% over logistic regression baseline** — derived via cross-validation and financial loss-function modeling tied to real stakeholder KPIs.

- **Deployed** a live Streamlit monitoring dashboard with SHAP explainability, real-time score feeds, and configurable alert thresholds — enabling **millisecond decision transparency** and CI/CD-ready ML production readiness.

CreditRisk Intelligence Engine — Fairness-Constrained ML | [GitHub ↗](#)

Python · XGBoost · SHAP · LIME · AIF360 · Streamlit

- **Architected** an end-to-end credit risk pipeline with ensemble stacking (XGBoost + Random Forest), achieving **0.91 ROC-AUC and 93% accuracy** via systematic hyperparameter tuning and k-fold cross-validation.
- **Enforced** fairness constraints using AIF360 (demographic parity + equal opportunity), achieving **99.99% fairness compliance** across all protected segments — fully aligned with EU AI Act guidelines for high-risk AI systems.

TruthGuard AI — Explainable Misinformation Detection | [GitHub ↗](#)

Python · NLP · TF-IDF · Ensemble (LR + RF + SVM) · LIME · Streamlit

- **Trained** a stacked ensemble classifier on **72,000+ articles** using TF-IDF bigram engineering — delivering high-speed misinformation classification with LIME per-prediction explanations for editorial transparency.

EDUCATION

B.Tech — Electronics & Communication Engineering

Expected May 2026

Management Education and Research Institute, Maharshi Dayanand University, New Delhi

Diploma in Data Science & Generative AI

March 2026

Boston Institute of Analytics (BIA)

Diploma in Artificial Intelligence

March 2026

Boston Institute of Analytics (BIA)

Diploma — Electronics & Communication Engineering (91%)

2023

Meerabai DSEU, Maharani Bagh Campus

CORE COMPETENCIES

Machine Learning (Supervised/Unsupervised) · Predictive Maintenance & Anomaly Detection · Industry 4.0 / IIoT Analytics · Time Series Forecasting · Explainable AI (SHAP, LIME, AIF360) · Feature Engineering · Natural Language Processing · Statistical Modeling & A/B Testing · Data Pipeline & ETL Development · PLC / SCADA / HMI Systems · Model Deployment (Streamlit, FastAPI) · Responsible & Fairness-Aware AI · Power BI & Tableau · SQL & PySpark · Root Cause Analysis · Generative AI & LLM Engineering · End-to-End ML Project Ownership · Cross-functional Stakeholder Communication

CERTIFICATIONS & ACHIEVEMENTS

- Data Analysis using SQL, Power BI, Tableau & Excel — Comprehensive Analytics & Visualization | Udemy
- LLM Engineering: Master AI and Large Language Models | Udemy
- Active Kaggle Contributor — participated in data science competitions focused on statistical modeling and ML best practices
- YouTube Educator — Founder of The Learning Lounge; created accessible Data Science & AI content for 1,000+ learners